

PRESSURIZED README

Project Title

PRESSURIZED

Creator

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Short Description

PRESSURIZED is a tactile arcade experience in which a player pilots a submarine through a chaotic underwater world by operating a massive hand-built panel of real switches, dials, levers, and buttons. The game is built in Unity and communicates in real time with a custom Arduino-based physical controller mounted in a hand-fabricated arcade cabinet.

How to Run

See Setup Instructions document for full step-by-step launch procedure. In brief:

1. Connect all three daisy-chained Arduino via single USB cable to the laptop.
2. Power on the Bluetooth speaker and pair to the laptop.
3. Power on the monitor and LED power supply via the power strip.
4. Launch the PRESSURIZED Unity build executable.
5. Verify serial connections and LED response. The game is ready.

Required Software

- Unity (standalone build. No Unity Editor required for play)
- Arduino IDE (for debugging only)

Required Hardware

- PRESSURIZED arcade cabinet
- PRESSURIZED control deck with 2x Arduino Mega + 1x Arduino Uno
- Laptop (MacBook or equivalent, build is for MacOS)
- Monitor (installed in cabinet)
- Bluetooth speaker
- Power strip

Folder Structure

01_ProjectBuild: Unity standalone build executable and required data folders

02_SourceFiles: Unity project source files, Arduino scripts

03_FinalDocumentation: Final Documentation, Artist Statement

04_ProcessDocumentation: Process Documentation

05_ProcessDocumentation_IMAGES: Development Timeline with photos

06_PresentationMaterials: Showcase poster, presentation description

07_SetupInstructions: Full installation and setup instructions

08_ReadMe: This document

Known Issues

- Arduino RAM at capacity on all three boards. New processing logic should not be added to microcontrollers.
- Input state inversions may occur on cold boot; restart the Unity build to resolve.
- LED system requires Unity to be running to receive animation data; standalone LED behavior is not implemented.